

MARQUETTE HARBOR, MI, USA

WMO: 997262

Lat: 46.546N

Lon: 87.379W

Elev: 188

StdP: 99.09

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.0	(c) -17.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 13.8	(k) -9.4	(l) 10.9	(m) -6.7	(n) 4.2	(o) 250	(p) 0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.1	(c) 27.5	(d) N/A	(e) 25.7	(f) N/A	(g) 24.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.0	(b) 9.0	(c) 7.6	(e) -22.2	(f) 31.3	(g) 2.8	(h) 1.8	(i) -24.2	(j) 32.6	(k) -25.8	(l) 33.7	(m) -27.4	(n) 34.7	(o) -29.4	(p) 36.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-7.4	-7.3	-2.2	3.5	8.9	14.0	18.7	18.5	14.7	8.3	1.5	-4.4		
	DBStd	10.33	5.69	5.29	5.50	4.39	4.60	4.23	3.68	3.05	3.96	4.41	5.00	4.70		
	HDD10.0	2489	540	486	380	201	77	9	0	0	6	87	258	445		
	HDD18.3	4751	798	719	637	445	293	141	41	36	121	312	506	703		
	CDD10.0	901	0	0	2	6	44	131	271	264	147	35	2	0		
	CDD18.3	121	0	0	0	0	2	12	53	41	12	2	0	0		
	CDH23.3	489	0	0	0	2	21	76	206	134	46	4	0	0		
	CDH26.7	84	0	0	0	0	4	16	37	19	8	1	0	0		
(14) Wind	WSAvg	3.1	3.7	3.5	3.5	3.5	3.3	2.5	2.4	2.5	2.8	3.0	3.3	3.4		
(15) Precipitation	PrecAvg	814	50	44	55	73	73	77	68	73	100	90	69	59		
	PrecMax	1141	168	136	145	165	196	180	137	299	210	208	185	124		
	PrecMin	341	0	0	0	0	0	0	0	0	9	23	0	9		
	PrecStd	157	30	25	32	37	41	38	37	45	45	42	35	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.1	16.7	19.7	26.3	29.0	30.4	29.4	27.8	23.4	15.0	6.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	3.3	3.0	10.8	15.1	22.2	25.9	27.3	26.8	24.2	19.6	11.9	4.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	1.6	1.1	7.6	12.3	19.0	22.8	25.6	24.9	21.9	16.4	10.0	3.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	0.4	-0.5	5.1	9.9	16.2	20.4	24.0	23.1	20.3	14.3	8.1	1.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		5.8	6.5	6.9	6.5	7.8	7.6	7.1	6.3	7.0	5.7	5.2	5.0		
	5% DB	MCDBR	6.3	8.0	10.6	11.2	13.1	11.4	9.5	8.8	9.5	9.1	7.3	6.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.281	0.315	0.342	0.378	0.379	0.386	0.372	0.360	0.331	0.302	0.267		
	taud		2.430	2.381	2.371	2.391	2.366	2.390	2.397	2.437	2.469	2.523	2.520	2.484		
	Ebn at Noon		843	900	913	915	888	886	874	874	854	832	795	807		
	Edh at Noon		67	87	103	111	118	116	114	105	93	76	63	58		
(44) All-Sky Solar Radiation	RadAvg		1.04	1.87	3.32	4.45	5.44	5.90	6.02	5.10	3.76	2.22	1.17	0.81		
	RadStd		0.08	0.25	0.32	0.47	0.39	0.45	0.32	0.33	0.25	0.26	0.15	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page